AI Meal Planner

Say what you crave. Get **one meal sized to you** and a shopping list.

Capstone project
by Mike
P1 pending



363 kcal
26 g protein
Older woman, losing weight



694 kcal
48 g protein
Active woman, maintaining



809 kcal
60 g protein
Average man, maintaining



850 kcal
63 g protein
Large athlete, gaining

1 The problem

Picking a meal is easy. Sizing it to you is not.

How much?
Daily budgets run from 1,200 to 4,521 kcal.

Is it right for me?
A limit can change the recipe or swap an ingredient.

What do I buy?
Even one pasta means five ingredients to price.

2 Four agents work like one kitchen

ORDER chicken fried rice, active woman, maintaining

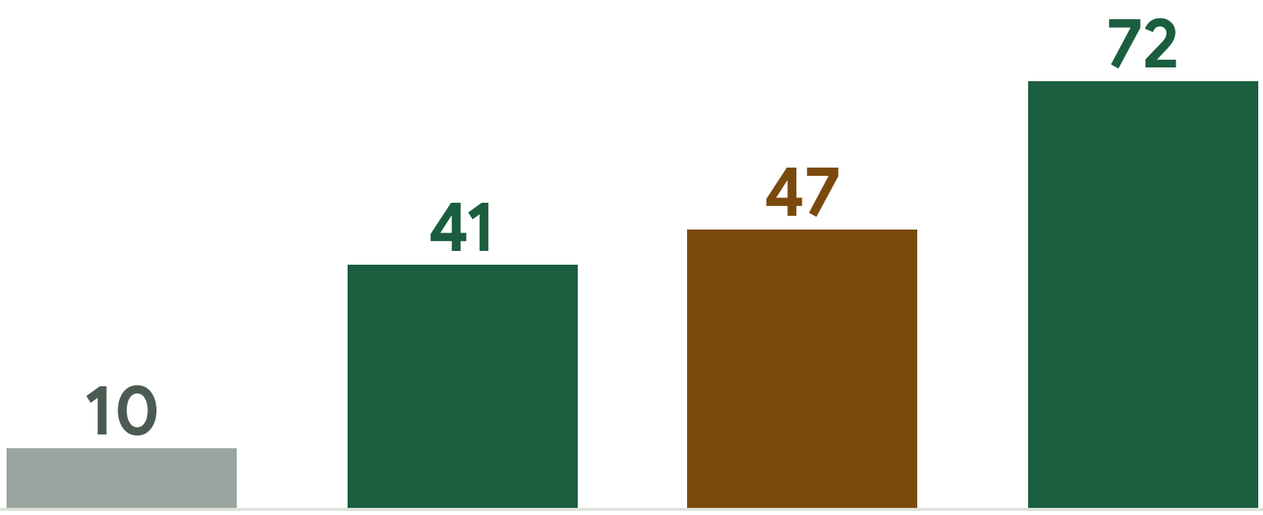
THE PORTION PLANNER
Calorie expenditure
Estimates a day's burn and sets the budget.
2,469 kcal budget

THE CHEF
Meal recommendation
Rules out conflicts, picks a recipe, sizes it.
691 kcal target

THE TASTER
Nutrition verification
Looks up each ingredient, labels estimates.
694 kcal, 3 of 6 estimated

THE SHOPPER
Supermarket
Matches items to store or category prices.
$24.50 estimated

3 Results: three fixes, from 10 to 72 of 72



First run
Before any fix

Portion balancing
Ingredients resized

Honest fallback
Off-menu cravings moved

Cooking fat, retry
Oil added to 6 recipes

0.4%
median calorie gap, worst 9.9%

5 of 5
guardrail cases hold

68 of 68
wider check, every recipe by name

Plans meeting calorie and macro targets on the project's own offline benchmark of 72 requests. 11 are fallback meals.

4 From body to meal target

Daily budget: 2,469 kcal — +125 from the model

2,344 kcal from body and activity

This meal: 28% of the day

691 kcal

The model scores R^2 0.996 on exercise calories, but adds only 3–10% of the day.

5 Hard limits first, then relevance

All recipes	34
Safe for a vegan	13
Relevant to craving	5
Served	1: Tofu Fried Rice

6 Where the 694 kcal come from

459 estimated, 66% | 235 reference

Only 3 of 6 ingredients are estimates, but they include the biggest items.

7 Architecture and tech stack

Streamlit
Live, public demo

FastAPI
Typed JSON, keys, rate limits

React
Guided flow, run locally

SQLite
Local history, via SQLModel

`MealPlanningService`

Calorie expenditure
HistGradientBoosting model

Meal recommendation
TF-IDF search, diet rules

Nutrition verification
Local tables; USDA, FatSecret with keys

Balance and reconcile
Linear programme for portions

Supermarket
Local store items and category prices

Built with: Python · Docker · GitHub Actions · Cloud Run, not yet deployed · OpenTelemetry · Gemini, optional

630 Python tests, 95% coverage

8 What it does not prove yet

Measured
Targets met offline, but checks guided the fixes.

Estimated
68 of 72 plans include an estimate. AMDR is a daily range used on one meal.

Not tested
No users, no clinical review, not deployed. Taste and health outcomes are unknown.

9 What comes next

NOW	NEXT	LATER
Demo on by default, migrations, deploy	USDA benchmark, real exercise data, allergen tags	75–100 recipes, live prices, testing with people